

Omics in Diabetes Epidemiology

What a Review Article Taught Me About Study Design

A critical analysis of “Lessons and Applications of Omics Research in Diabetes Epidemiology”

Yu, Tam, Huang, Shi, Lim, Chan & Ma (2024), Current Diabetes Reports

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Table of Contents

1. Research Background.....	3
2. Paper Overview.....	3
3. Methodological Framework.....	3
4. Key Findings.....	4
4.1 Findings by Omics Layer.....	4
5. Critical Analysis.....	5
5.1 Strengths.....	5
5.2 Weaknesses.....	5
5.3 Advantages vs. Limitations, at a Glance.....	5
5.4 Unitary PRS vs. Partitioned / Clustered PRS.....	6
5.5 My Perspective.....	6
6. Limitations.....	6
6.1 Single-Omics vs. Integrated Multi-Omics Study Design.....	7
7. Future Directions.....	7
8. Lessons Learned.....	7
9. Potential Research Extensions.....	8
10. Questions I Still Have.....	8
11. How This Paper Influences My Future Projects.....	8
12. Conclusion.....	9
References.....	9

1. Research Background

Diabetes now affects somewhere around 420 million people worldwide, and that number is the starting point the authors use to justify the whole review. Classic epidemiology — the population- and family-based kind — has done a lot of the groundwork here: it told us how diabetes clusters in families, how it varies by place, and it generated plenty of hypotheses about causes. What it couldn't really do on its own is explain mechanism. That's the gap this review sits in, and it's a slightly different gap than the one I looked at last time.

The DeepInsight review I read before this one was about a computational trick — reshaping tabular data so an existing architecture (a CNN) could be pointed at it. This review isn't about a trick at all. It's about infrastructure and planning. The argument, as I read it, is that most of the value of omics data doesn't come from a clever downstream model — it comes from whether the cohort was set up, sampled, and biobanked in a way that makes the omics data usable in the first place. That's a much less flashy claim, but by the end of the paper I was fairly convinced it's the more consequential one for anyone actually running a study rather than just publishing a method on top of someone else's dataset.

2. Paper Overview

Paper reviewed: “Lessons and Applications of Omics Research in Diabetes Epidemiology” (2024), Gechang Yu, Henry C. H. Tam, Chuiguo Huang, Mai Shi, Cadmon K. P. Lim, Juliana C. N. Chan & Ronald C. W. Ma, *Current Diabetes Reports*.

Type: Review article. Area: Diabetes epidemiology, genomics, epigenomics, proteomics, metabolomics.

The review walks through the major omics layers one at a time — genetics and polygenic risk scores, epigenetics (methylation, miRNA, histone marks, telomere length), proteomics, and metabolomics — and closes with a section on integrating them. Throughout, the authors lean heavily on their own cohorts, the Hong Kong Diabetes Register and Hong Kong Diabetes Biobank, to illustrate what actually incorporating omics into an epidemiological study looks like in practice, rather than just summarizing what other groups have published.

3. Methodological Framework

There's no single pipeline to diagram here the way there was for the DeepInsight papers, and I don't think it would be honest to force one. What the review does instead is walk each omics layer through roughly the same arc: how the measurement technology evolved, what's been discovered with it in diabetes specifically, and what still limits its clinical usefulness. Genetics goes from linkage analysis, to candidate genes, to GWAS, to sequencing, to summary risk scores. Epigenetics moves from single-locus methylation studies to genome-wide EWAS to multi-omic integration with genotype data. The throughline across all of them is a study-design question that keeps resurfacing: what do you need to have collected, and how, before any of this analysis is even possible.

That's also where Table 1 in the original paper does a lot of quiet work. It's not a results table, it's a planning table — sample type needed, typical sample size, and “special considerations” for each kind of biomarker. Things like: DNA quality matters less for common-variant genotyping than for other applications; methylation profiling needs cell-type composition in blood accounted for; miRNA profiling is

ruined by lithium heparin collection tubes and affected by haemolysis. I don't think I've read a methods table before that was this explicitly about logistics rather than statistics, and it's probably the single most reusable part of the paper.

4. Key Findings

A handful of numbers from across the different sections stuck with me enough to write down:

- The largest T2D GWAS to date is a multi-ancestry meta-analysis of over 2.5 million individuals (including ~0.4 million cases), identifying 1,289 distinct association signals across 611 loci — most of them common variants with small effect sizes (odds ratios under 1.1).
- An exome sequencing study of over 20,000 T2D cases found that the strongest rare-variant gene-level signals explained only 25% of the heritability captured by the strongest common-variant signals — which runs against the common assumption that rare variants are where the “missing heritability” has been hiding.
- A 67-variant T1D genetic risk score (GRS-2) discriminated T1D from T2D and controls with an AUC of 0.93, up from 0.88 for the earlier 30-variant version.
- In UK Biobank, individuals in the top 2.5% of a T2D polygenic risk score distribution had a 9.4-fold higher risk than those in the bottom 2.5%; a trans-ancestry PRS (PRS-CSx) found a comparable 2.5–4.5-fold risk increase for the top 2% across European, African, Hispanic/Latino, and East Asian ancestry groups.
- A DNA methylation site in TXNIP was independently flagged as associated with incident T2D in one cohort (LOLIPOP) and then, separately, showed up again in the DCCT/EDIC metabolic-memory analysis of T1D complications — two very different study designs converging on the same locus.
- NMR metabolomic profiling in the China Kadoorie Biobank identified 163 metabolites associated with T2D risk, 147 of which remained significant after adjusting for baseline glucose.

4.1 Findings by Omics Layer

Layer	Representative Finding	What It Adds
Genetics / GWAS	611 loci, 1,289 signals from a 2.5M-person meta-analysis	Defines the common-variant landscape of T2D risk
Sequencing (rare variants)	Rare-variant signals explain ~25% of common-variant heritability	Tempers assumptions about where missing heritability sits
Polygenic Risk Scores	T1D GRS-2: AUC 0.93; trans-ancestry T2D PRS: 2.5–4.5-fold risk gradient	Aggregates GWAS signal into an individual-level score
Genetic clustering	“Palette model”: 5–10 T2D genetic subtypes (Udler et al.)	Reframes T2D as heterogeneous rather than one disease
Epigenetics (methylation)	TXNIP locus replicated across independent cohort designs	Links environmental exposure to gene regulation
Epigenetics (miRNA)	miR-122-5p / miR-455-3p flag liver cancer risk in T2D patients	Circulating biomarker candidates from routine serum

Layer	Representative Finding	What It Adds
Proteomics	MR identifies SIRPG, IL27-EBI3, CTRB1 as candidate T1D drug targets	Distinguishes causal proteins from incidental correlates
Metabolomics	163 T2D-associated metabolites in China Kadoorie Biobank	Captures dynamic, diet- and microbiome-linked risk signal

What I kept noticing across this table is how often Mendelian randomization shows up as the tool doing the actual argumentative work — not the omics assay itself. The assays generate candidate associations; MR is what the authors use, over and over, to decide whether a candidate is worth taking seriously as causal rather than just correlated. That's a pattern I didn't consciously register the first few times it appeared, and only noticed once I saw it recur in the proteomics, methylation-QTL, and metabolomics sections independently.

5. Critical Analysis

5.1 Strengths

- Table 1 is a genuinely practical, reusable planning resource, not a rehash of results
- Repeated, consistent use of Mendelian randomization to separate causal biomarkers from mere correlates
- Honest inclusion of a counter-intuitive finding (rare variants explaining less heritability than expected) rather than only reporting confirmatory results
- Cross-cohort replication (TXNIP) is highlighted as a marker of a credible signal, not just asserted

5.2 Weaknesses

- Heavy reliance on the authors' own cohort (HKDR/HKDB) for illustrative examples
- Breadth across five omics layers comes at the cost of depth in any one of them
- Not a systematic review, so there's no stated process for how examples were selected
- PRS transferability across ancestries is mentioned repeatedly but never quantified in one place

5.3 Advantages vs. Limitations, at a Glance

Advantage	Corresponding Limitation
Grounded in real cohort-building experience (HKDR/HKDB)	Examples skew toward one research group's own studies
Table 1 gives concrete, actionable sample-handling guidance	Guidance is blood-sample-centric; other tissue types get little space
MR used consistently to test causality, not just association	MR's own assumptions (no pleiotropy, valid instruments) go unstated
Covers genetics through metabolomics in one document	Each layer gets a few paragraphs rather than a full treatment

5.4 Unitary Polygenic Risk Scores vs. Partitioned / Clustered PRS

The comparison I kept circling back to while reading wasn't between two computational architectures — it was between two ways of summarizing the same genetic information. A standard PRS collapses thousands of variant effects into a single number. The “palette model” that Udler and colleagues developed instead decomposes those same variant effects into five (later ten) genetic clusters, each associated with a different metabolic and clinical pattern. The review treats this mostly as a promising extension; I think it's worth laying the two approaches side by side.

Aspect	Unitary PRS	Clustered / Partitioned PRS
What it represents	A single weighted sum of variant effects	Multiple sub-scores, each tied to a distinct genetic mechanism
Clinical interpretability	One risk number, no mechanistic detail	Each cluster suggests a candidate pathway (e.g. beta-cell, insulin resistance)
Data requirements	Works from standard GWAS summary statistics	Needs enough associated loci and traits to support clustering
Validation status	Widely validated across large biobanks	Newer; five-cluster and ten-cluster versions already disagree with each other
Ancestry transferability	Already a known weak point (mostly European-derived)	Unclear — clustering itself was done in largely European discovery data

5.5 My Perspective

The part of this section I'm most skeptical about is exactly the part the review seems most enthusiastic about: the clustering. Genetic clustering methods are, at bottom, statistical groupings of correlated variant effects across a set of traits — and the fact that the five-cluster version got extended into a ten-cluster version in later work, with the clusters not being identical, tells me the boundaries between “types” of T2D genetic risk are still somewhat a function of how the clustering was run, not a fixed biological fact waiting to be uncovered. That doesn't mean the approach is wrong. It means I'd want to see the same clustering procedure applied independently to a non-European discovery cohort and check whether the same five-to-ten clusters fall out, before treating them as established disease subtypes rather than a useful but provisional way of organizing the data.

This is basically the same worry I had reading the DeepInsight review, just showing up in a different place. There, the concern was whether spatial proximity in a t-SNE-generated image reflected real biology or just an artifact of that particular projection. Here, the concern is whether cluster membership in a soft-clustering PRS decomposition reflects a real pathophysiological subtype or just an artifact of that particular clustering run on that particular discovery dataset. Both are cases where a data-driven grouping is doing a lot of interpretive work, and in both cases I think the grouping needs to be shown stable across independent runs and independent populations before it earns the label “mechanism.”

6. Limitations

The authors are upfront about several open problems: benchmarking and definitions are inconsistent across diabetic-complication genetics studies (they specifically flag diabetic kidney disease here), sequencing studies need sample sizes in the millions to reach exome-wide significance for validated

targets, and most T1D and T2D PRS work still skews toward European-ancestry discovery cohorts. They're also honest that PRS alone tends to perform poorly without being combined with existing clinical risk factors like BMI, GADA, and fasting glucose.

6.1 Single-Omics vs. Integrated Multi-Omics Study Design

Single-Omics Design	Integrated Multi-Omics Design
One layer measured per cohort or sub-study	Genome, epigenome, proteome, metabolome measured in the same individuals
Findings compared across separate cohorts after the fact	pQTL / meQTL analyses link layers directly within one dataset
Cheaper, faster to run	Requires far more storage, linkage, and analytic infrastructure
Easier to interpret in isolation	Needs multidisciplinary teams to interpret jointly

7. Future Directions

The authors expect the field to keep moving toward measuring multiple omics layers in the same individuals rather than stitching together separate cohort-level findings, expanding into layers this review didn't have space for (radiomics, gut microbiome, the exposome), and building out the machine learning methods needed to actually integrate that much heterogeneous data. They're also clear that none of this works without solving fairly unglamorous infrastructure problems first — data storage costs, data linkage across platforms, and getting the right mix of clinicians, epidemiologists, geneticists, and data scientists in the same room.

8. Lessons Learned

The heritability finding was the one that actually changed how I think about a debate I thought I already understood. I'd absorbed the general narrative that rare variants were probably where the “missing heritability” of complex traits like T2D was hiding, mostly because that's the intuitive place to look once common variants have been exhausted. Seeing a direct sequencing study measure that and come back with rare variants explaining only a quarter of what the common-variant signals already explain was a useful correction — a reminder that an intuitive-sounding explanation still needs to actually be measured before I treat it as the answer.

I also hadn't previously thought much about polygenic risk scores as products that need other ingredients to work. The review is fairly blunt that a PRS by itself is often not that useful clinically, and only becomes actionable combined with BMI, autoantibody status, or fasting glucose. That reframes PRS less as a standalone diagnostic and more as one input feeding into a larger risk model, which changes how I'd think about evaluating any future PRS-based method I read about.

And the practical side of this paper — Table 1, the sample-handling notes, the biobanking advice — pushed something up my list that I don't think I'd weighted properly before: how much of the eventual quality of an omics analysis is determined months or years earlier, at the point of sample collection, before any model gets trained on it at all.

9. Potential Research Extensions

- Run the same soft-clustering PRS procedure on a non-European T2D GWAS and check whether the same cluster structure emerges
- Build a small Table-1-style sample-planning checklist for a hypothetical cohort study, as a way of actually using the framework rather than just reading about it
- Pick one biomarker from the proteomics section and try replicating the Mendelian randomization logic using public GWAS summary statistics
- Compare the quantitative size of the ancestry gap in T2D PRS performance directly, since the review mentions it in several places but never puts one number on it

10. Questions I Still Have

- Would the five- or ten-cluster “palette” structure hold up if the clustering were re-run from scratch on an East Asian or African-ancestry T2D GWAS, or is it partly an artifact of the discovery cohort it was built from?
- Is the 25%-of-heritability figure for rare variants likely to move much as sequencing studies scale toward the million-sample mark the authors say is needed, or is that ceiling itself uncertain?
- When multi-omics layers are measured in the same individuals rather than compiled across separate cohorts, how much does that actually change the biological conclusions versus just making the pQTL/meQTL analyses more convenient to run?
- Given how much of this review draws on the Hong Kong Diabetes Register and Biobank, how different would the picture look reading the equivalent review written from a European- or African-ancestry-focused diabetes consortium?

11. How This Paper Influences My Future Projects

This pushed me toward wanting to actually build the kind of planning document Table 1 is, rather than just admiring it — a short checklist I'd fill out before collecting any biospecimens for a future project, covering sample type, expected sample size, and the handling quirks specific to whatever omics layer I'm targeting. I'd also like to make a habit of running a quick Mendelian randomization sanity check any time I'm tempted to call a biomarker “causal” rather than just “associated,” since that distinction did a lot of quiet work throughout this review and I don't think I've been rigorous enough about it in my own reading up to now.

More broadly, this shifted how much weight I put on study design versus modeling choices when I evaluate a paper. Reading the DeepInsight review, the interesting decisions were all downstream, in how the data was represented and modeled. Reading this one, the interesting decisions are almost all upstream, in how the cohort was built and what was collected before anyone ran an analysis at all. I want to hold both of those lenses at once going forward, instead of defaulting to the modeling one just because it's the flashier half of the pipeline.

12. Conclusion

This review isn't selling a new algorithm or a new biomarker — its real argument is that the design of the cohort and the discipline of the biobanking matter as much as anything that happens downstream in analysis. That's a less exciting claim to make than “we found a new gene,” but by the end of the paper I found it the more durable one. The rare-variant heritability result and the PRS-needs-other-inputs point both suggest that a lot of the field's remaining problems won't be solved just by getting bigger sample sizes on the same designs — they need deliberate upfront decisions about what gets collected, from whom, and how it's stored. Whether the palette-model style of genetic clustering turns out to be real biological subtyping or a useful-but-provisional statistical convenience is still an open question, and one this review is honest enough to leave open rather than oversell.

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